

Sandip Sapkota

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SUMMARY

BSc.IT (Hons) student with hands-on experience building full-stack applications, AI-integrated projects, and Python programs. Comfortable across web development, database concepts, and beginner-level .NET, with a strong interest in software development and applied AI. Experienced in organizing technical events and collaborating in team-based academic projects.

PROJECTS

YatraNepal – Real-Time Public Transit Tracker

Solo Project

React, TypeScript, Node.js, PostgreSQL, WebSocket, Leaflet.js

- Built a real-time public transportation tracking app for Kathmandu Valley, broadcasting live bus and minibus positions over WebSocket every 3 seconds along real road geometry via OSRM
- Implemented interactive map layers (OpenStreetMap and satellite), location search via Nominatim geocoding, and route filtering using React, TypeScript, and Leaflet.js
- Designed a Node.js/Express backend with PostgreSQL (Drizzle ORM) to serve route data, calculate real-time ETAs using Haversine distance, and trigger proximity-based notifications
- Integrated traffic congestion overlays and modeled four real transit operators (Sajha Yatayat, Mahanagar Yatayat, Nepal Yatayat, Bagmati Yatayat)

Esports Championship Tournament

Academic Group Project

Mini Militia, PUBG

- Served as tournament organizer and coordinator for a college-level esports event featuring Mini Militia and PUBG competitions
- Coordinated scheduling, rules, and logistics across multiple teams, and led production of the formal project report and documentation

Tic-Tac-Toe with AI Opponent

Personal Project

Python, Minimax Algorithm

- Built a Python implementation of Tic-Tac-Toe featuring an unbeatable AI opponent using the minimax algorithm
- Applied game theory and recursive decision-tree search to evaluate optimal moves

EDUCATION

Techspire College

BSc.IT (Hons)

Kathmandu, Nepal
Sept 2025 – Sept 2028 (Expected)

St. Lawrence College

+2 Science, NEB – GPA: 3.81

Kathmandu, Nepal
2024 – 2025

CERTIFICATIONS

- CS50: Introduction to Programming with Python – Harvard University (HarvardX)
- Design Thinking – Training of Trainers

TECHNICAL SKILLS

Programming & Frameworks: Python, C# (.NET, beginner), HTML, CSS, JavaScript

Web Development: React, Node.js/Express, REST APIs

Database Concepts: MySQL, PostgreSQL, schema design

Other: Network Troubleshooting, Graphic Design, Git/GitHub